



École Polytechnique Fédérale de Lausanne (EPFL)

ACOUSTIC OFDM TRANSMISSION SYSTEM

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1 Introduction

Orthogonal Frequency Division Multiplexing (OFDM) is at the basis of modern wideband communication systems, including Wi-Fi and 4G/5G cellular networks. Its primary advantage lies in its ability to cope with severe frequency selective fading without complex time domain equalization. By splitting a high-rate data stream into many lower rate sub-streams transmitted on orthogonal subcarriers, we effectively slice the complex frequency response into narrow segments. Within each segment, we can approximate the channel as experiencing flat fading, assuming the distortion is constant across that tiny slice of bandwidth. This simplification allows the receiver to recover the data easily by applying a single complex correction factor (amplitude and phase) to each subcarrier, rather than using complex filters. In this project, we implemented a complete baseband OFDM transmitter and receiver in MATLAB and interfaced it with audio hardware to transmit data over the air using sound waves. The primary goal of this project was to gain a practical understanding of the entire digital communication chain, implementing every stage from the initial bit mapping down to synchronization and equalization. Beyond simply building the transceiver, we focused on analyzing its performance limits, specifically measuring how the Bit Error Rate (BER) evolves with respect to Signal-to-Noise Ratio (SNR) in both standard AWGN and challenging fading conditions. A significant portion of the work involved characterizing unknown acoustic environments using Power Delay Profiles and frequency response analysis, which informed our comparison of Block versus Comb channel estimation techniques for handling time varying channels. Finally, we tested the architecture to determine the maximum achievable spectral efficiency and throughput before the BER degraded.

2 System Implementation

2.1 Transmitter Architecture

The transmitter generates a complex baseband signal which is then upconverted to the acoustic passband. The processing chain is in Figure 1.

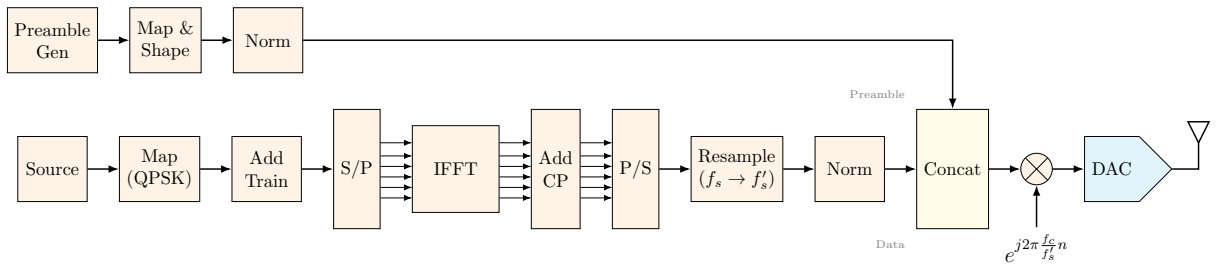


Figure 1: Acoustic OFDM Transmitter

2.1.1 Data Generation and Mapping

The process begins with generating a stream of binary data. This can be a random bitstream or encoded image data.

- Mapping: The bits are mapped to complex symbols using Quadrature Phase Shift Keying (QPSK).

- Serial to Parallel (S/P): The serial stream of QPSK symbols is grouped into blocks of length $N_{sc} = 512$, corresponding to the number of subcarriers.

2.1.2 OFDM Modulation (IFFT)

The core of the transmitter is the Inverse Fast Fourier Transform (IFFT). The parallel block of N_{sc} frequency-domain symbols is transformed into the time domain:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi \frac{nk}{N}} \quad (1)$$

This operation effectively modulates the data onto orthogonal subcarriers efficiently.

2.1.3 Cyclic Prefix (CP) Insertion

To mitigate Inter-Symbol Interference (ISI) caused by the multipath acoustic channel, a Cyclic Prefix is inserted at the beginning of each OFDM symbol. The last N_{cp} samples of the IFFT output are copied and added at the beginning of the symbol. For our default configuration, we used a conservative length of $N_{cp} = 256$ samples. This ensures that the guard interval is long enough for most indoor acoustic environments.

2.1.4 Preamble Generation

Independent of the OFDM chain, a synchronization preamble is generated. We used a Linear Feedback Shift Register (LFSR) to generate a pseudo-random single-carrier sequence. Then we shaped the preamble with a Root-Raised Cosine (RRC) filter. The preamble is essential for the receiver to detect the start of the packet (Frame Synchronization).

2.1.5 Digital Upconversion and Transmission

Since the baseband OFDM parameters differ from the hardware sampling rate ($f'_s = 48$ kHz), the discrete-time OFDM signal is resampled from the symbol rate to the hardware sampling rate. To ensure consistent energy levels between the synchronization and payload segments, both the generated preamble and the resampled OFDM data are individually normalized. They are then concatenated in the time domain, with the preamble placed before the data payload. The combined baseband signal is digitally upconverted by mixing it with a carrier frequency f_c (default 8 kHz) to shift it into the audible passband of the loudspeaker:

$$s_{tx}[n] = \text{Re} \left\{ x[n] \cdot e^{j2\pi \frac{f_c}{f'_s} n} \right\} \quad (2)$$

Finally, the total signal is normalized once more to fit within the dynamic range of $[-1, 1]$. This step is critical to prevent clipping at the Digital-to-Analog Converter (DAC) during the actual acoustic transmission.

2.2 Receiver Architecture

The receiver (shown in Figure 2) performs the inverse operations, with critical additions for synchronization and equalization.

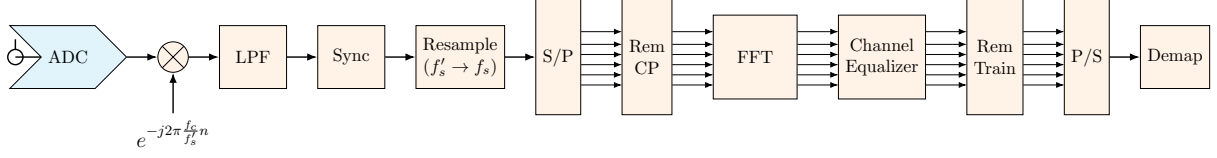


Figure 2: Acoustic OFDM Receiver

2.2.1 Frame Synchronization

Reliable communication depends critically on the receiver's ability to identify the precise start of the packet before any demodulation can occur. We implemented a robust frame synchronization algorithm utilizing the known single carrier preamble sequence. The structure of this synchronization is illustrated in Figure 3.

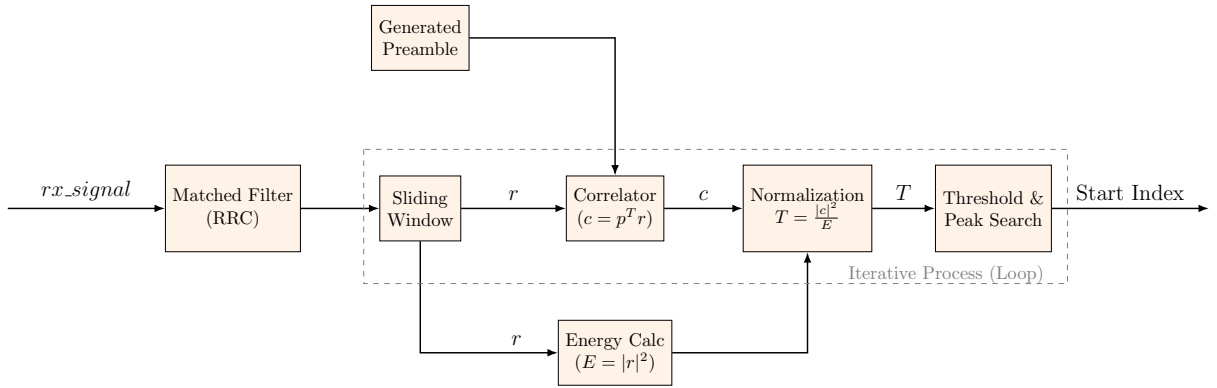


Figure 3: Frame Synchronization Algorithm Block Diagram

The incoming baseband signal is continuously fed into a sliding window correlator that compares the live stream with a copy of the preamble. Since the absolute amplitude of the received signal fluctuates significantly depending on the distance between the speaker and microphone, relying solely on the raw correlation peak is unreliable. To address this, the system simultaneously calculates the energy of the signal within the current window. The correlation output is then normalized by this energy value. A peak search algorithm identifies the sample index where the normalized correlation exceeds a predefined threshold as the start of the OFDM frame.

2.2.2 Demodulation and Equalization

Once the frame start is established, the receiver extracts the OFDM symbols. The cyclic prefix is stripped from the beginning of each symbol block to remove samples contaminated by Inter-Symbol Interference from previous reflections. The remaining time domain samples are converted back to the frequency domain using the Fast Fourier Transform (FFT). These recovered subcarriers are then passed to the channel equalization block, which estimates and compensates for the channel's frequency response (as detailed in the Channel Estimation section), before finally being demapped to recover the original binary data.

3 Channel Estimation and Equalization

Following synchronization and the transformation to the frequency domain via the FFT, the received symbols $Y[k]$ are distorted versions of the transmitted symbols $X[k]$. In a multipath acoustic environment, the channel introduces frequency selective fading, represented by the complex channel frequency response $H[k]$. The relationship at the receiver is given by:

$$Y[k] = H[k] \cdot X[k] + W[k] \quad (3)$$

where $W[k]$ represents the additive white Gaussian noise. To recover the original data, the receiver must first accurately estimate the channel response $\hat{H}[k]$ and then perform equalization. For this system, we implemented an equalizer, which compensates for the channel effects by dividing the received signal by the estimate:

$$\hat{X}[k] = \frac{Y[k]}{\hat{H}[k]} \quad (4)$$

The critical challenge in this project was implementing a channel estimation strategy capable of coping with the dynamic nature of the acoustic environment. We first implemented the Block Estimation and later the Comb Estimation for time varying channels.

3.1 Block Estimation and Phase Tracking

The simplest approach, Block Estimation, relies on the assumption that the channel is slowly fading and remains quasi static over the duration of a transmission packet. In this scheme, a full OFDM training symbol (where every subcarrier is a known pilot $P[k]$) is positioned at the beginning of the data payload.

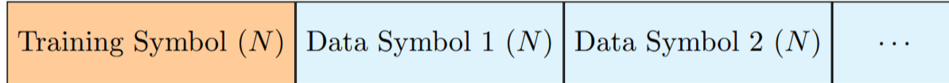


Figure 4: Serial stream structure for Block Estimation.

As shown in Figure 4, the training symbol effectively acts as a reference preamble. The receiver estimates the channel response once at the beginning of the frame using this known symbol. This single estimate is then applied to equalize all subsequent data symbols in the packet.

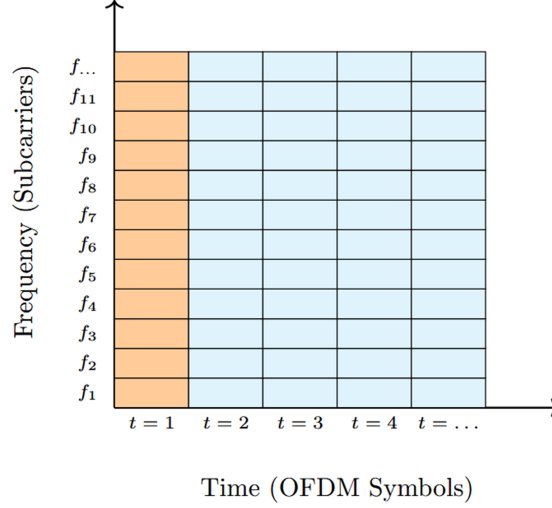


Figure 5: Time-Frequency grid representation.

Figure 5 illustrates the operation in the Time-Frequency domain. The channel is measured at $t = 1$ across all frequencies. For the remainder of the packet ($t = 2, 3, \dots$), the receiver assumes the channel H has not changed. While spectrally efficient (low overhead), this method fails if the channel changes during the transmission.

To mitigate phase drift without introducing more pilots, we attempted to implement a Viterbi-Viterbi phase tracking algorithm as described in the Section 5.

3.2 Comb Estimation

To address the failure of the block method in dynamic environments, we implemented Comb Estimation. This technique abandons the static channel assumption in favor of continuous tracking.

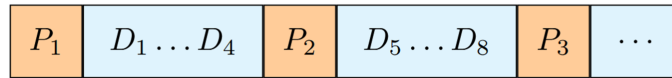


Figure 6: Serial stream structure for Comb Estimation.

As illustrated in Figure 6, instead of a single preamble, known pilot tones are inserted into the data stream at regular intervals. For our implementation, we selected a pilot spacing of $L = 5$, meaning one pilot is inserted for every four data subcarriers ($P_1, D_1 \dots D_4, P_2 \dots$).

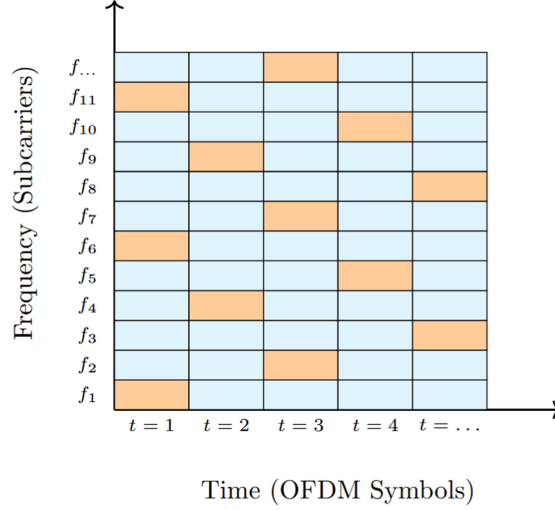


Figure 7: Time-Frequency grid representation.

This arrangement creates a pattern in the time-frequency grid, as shown in Figure 7. The receiver extracts the received signals at these specific pilot locations to obtain instantaneous channel estimates. To recover the channel response for the data subcarriers situated between the pilots (the blue cells), we applied linear interpolation across the frequency domain. By repeating this process for every OFDM symbol ($t = 1, t = 2, \dots$), the Comb method generates new channel estimates during the whole transmission, allowing it to track rapid phase rotations and amplitude variations.

3.3 Performance Comparison

The necessity of Comb estimation for acoustic channels is clearly demonstrated by the experimental results. Figure 8 presents a side-by-side comparison of the received constellations of Channel 2 transmission using Comb estimation (left) and Block + Viterbi estimation (right).

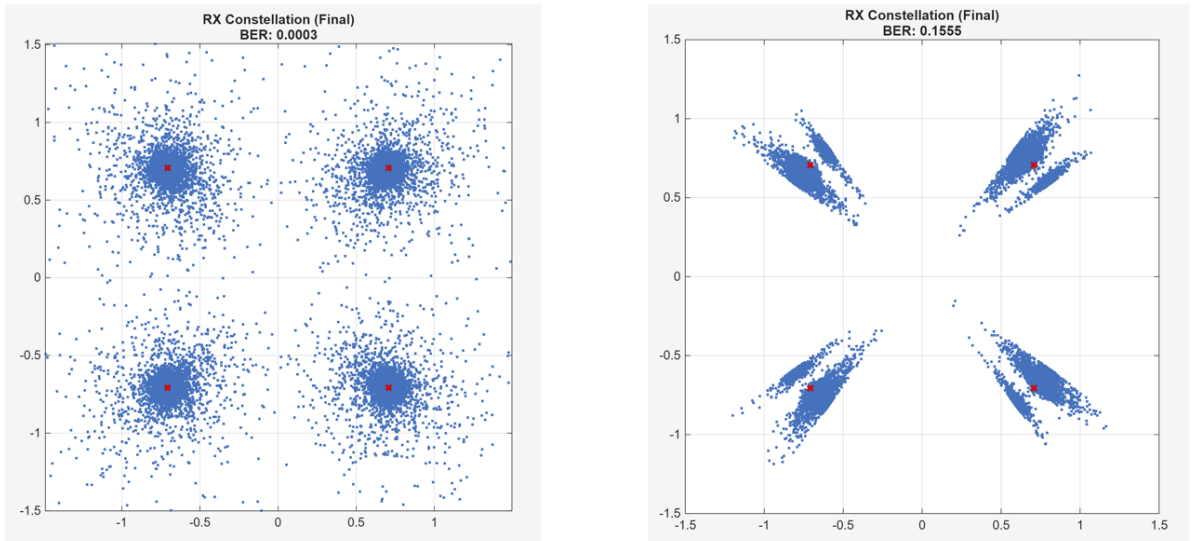


Figure 8: Comparison of equalization performance.

The Block + Viterbi estimation approach results in a degraded constellation with a Bit Error Rate of 15.5%, as the outdated channel estimate fails to correct the continuous phase rotation. In contrast, the Comb estimation method shows a stable, compact constellation with a BER of nearly 0%. The ability to continuously track the channel state ensures the robustness required for practical acoustic transmission.

4 Performance on AWGN Channel

The first phase of our project focused on validating the complete transmission chain using a standard Additive White Gaussian Noise (AWGN) channel (Channel 1). This step was to verify the correctness of our system.

As this was one of the initial tasks, we used Block channel estimation. We wanted to measure up to which SNR we could get a BER lower than 1%. The BER was measured against the SNR by varying the noise power in the emulator.

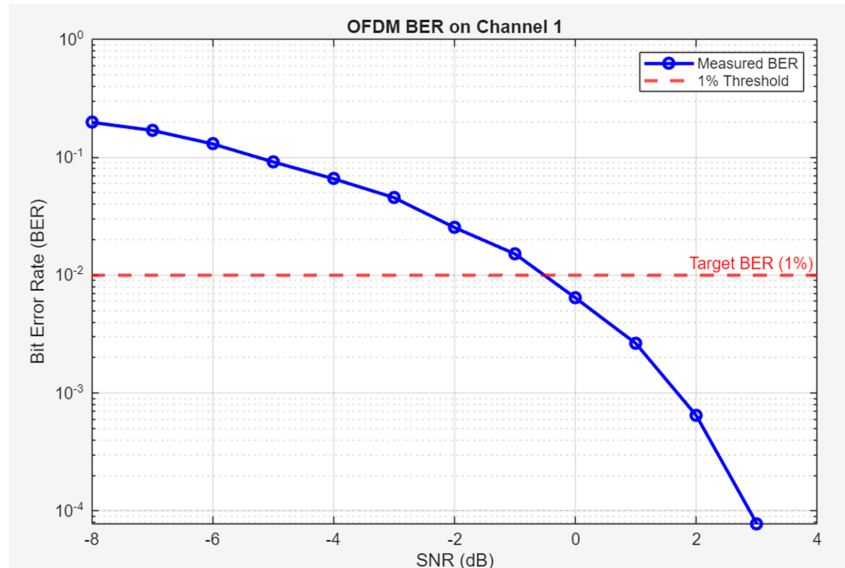
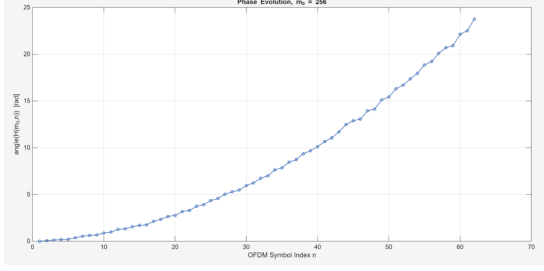


Figure 9: BER vs SNR for Channel 1 using Comb Estimation.

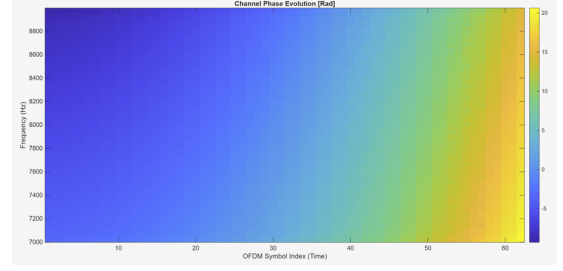
As shown in Figure 9, we achieved a BER < 1% at an SNR of approximately 0 dB (at -1 dB the BER exceeded > 1%).

5 Phase Tracking on Time-Varying Channels

Channel 2 introduced a time-varying phase rotation, as shown in Figure 10, simulating a receiver with accelerated motion. This phenomenon, as expected, caused the Block Estimation used previously on static channels to fail after the transmission of a few OFDM symbols, obtaining a final BER $\approx 46\%$. The effect of the phase drift introduced by the channel is clearly visible in Figure 11.



(a) Single Subcarrier



(b) All Subcarriers

Figure 10: Channel 2 Phase Evolution.

To handle this, we first implemented Block Estimation with Viterbi tracking, and later Comb Estimation as seen in Figure 8.

5.1 Block Estimation with Viterbi-Viterbi algorithm

We implemented a phase tracking algorithm (Viterbi-Viterbi algorithm) that raises the received symbols to the 4th power (for QPSK) to remove modulation and estimate the phase error.

$$\hat{\theta} = \frac{1}{4} \arg(-r[k]^4) \quad (5)$$

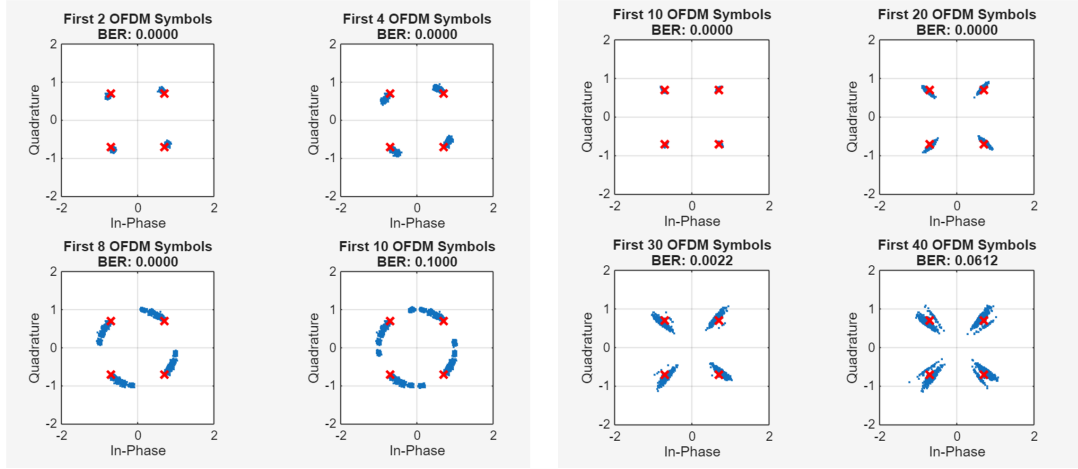


Figure 11: Constellation evolution with Block and Block + Viterbi tracking.

As seen in Figure 11, the tracker works well initially. However, around symbol 40, the BER spikes to 6.12%. This failure is due to the fact that the Viterbi algorithm has a $\pi/2$ phase ambiguity. If the noise plus the accelerating rotation exceeds $\pm 45^\circ$ between two estimates, the tracker snaps to the wrong quadrant. Since our implementation tracks subcarriers independently, random noise causes different subcarriers to slip at different times across a time span of five OFDM symbols approximately.

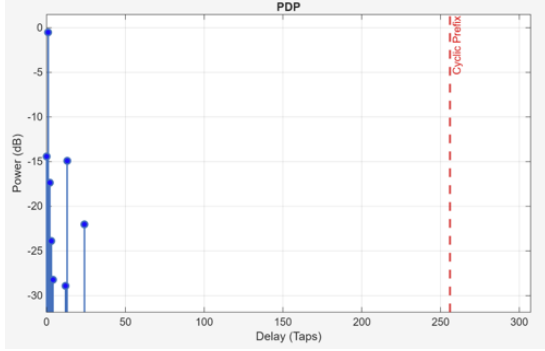
6 Channel Characterization

After evaluating our transmission system on Channel 1 and 2, we analyzed Channels 3, 4, and 5 using the calculated Power Delay Profile (PDP), Frequency Response and observing

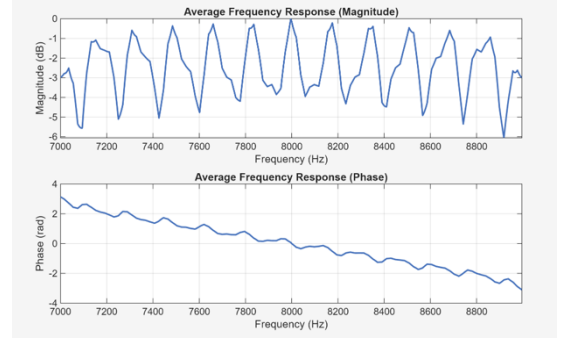
channel evolution over time.

6.1 Channel 3: Multipath

Channel 3 exhibited a longer Power Delay Profile compared to the ones we observed for Channel 1 and 2.



(a) PDP



(b) Frequency Response

Figure 12: Channel 3 Characteristics.

Channel 3 was simulating a multipath environment (or a channel subject to echo) as shown by the three distinct taps in the Power Delay Profile. Despite the long delay spread, our Cyclic Prefix ($N_{CP} = 256$) was sufficient to prevent ISI, as the PDP energy decayed before the CP limit. The magnitude and the phase of Channel 3 were static over time.

6.2 Channel 4: Band-Stop Filter

Channel 4 was also static, but its frequency response showed a deep attenuation notch, destroying specific subcarriers in the middle part of the bandwidth, making it a stop-band filter.

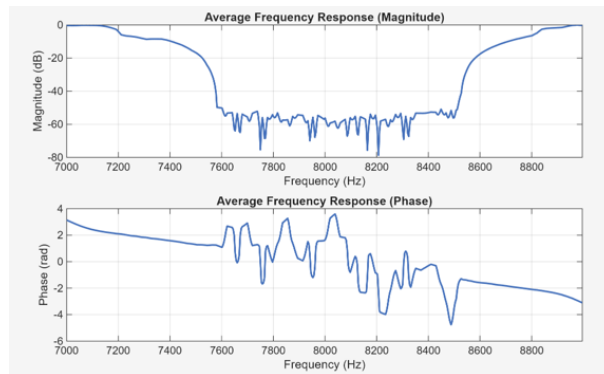
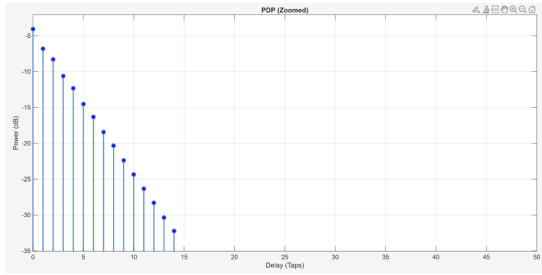


Figure 13: Channel 4 Frequency Response

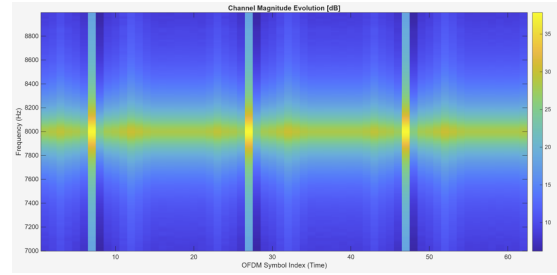
This resulted in a BER of 25% because the subcarriers located in the stop-band (which occupied approximately half of the total bandwidth) were completely lost to noise, causing the system to "random guess", while the others were recovered correctly.

6.3 Channel 5: Time-Varying Magnitude

Channel 5 introduced periodic impulses in the channel magnitude evolution. This represents a channel that "blinks" rhythmically. Comb pilots were essential here to track the amplitude variations during the transmission. The Power Delay Profile for Channel 5 displays an exponentially decaying multipath structure, dominated by a strong initial path followed by dense, decaying reflections. Unlike Channel 3, which is characterized by distinct, delayed echo taps, the delay spread in Channel 5 is more contained.



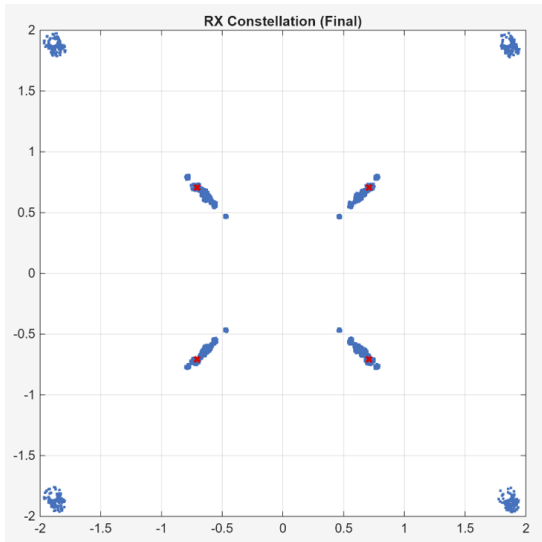
(a) PDP



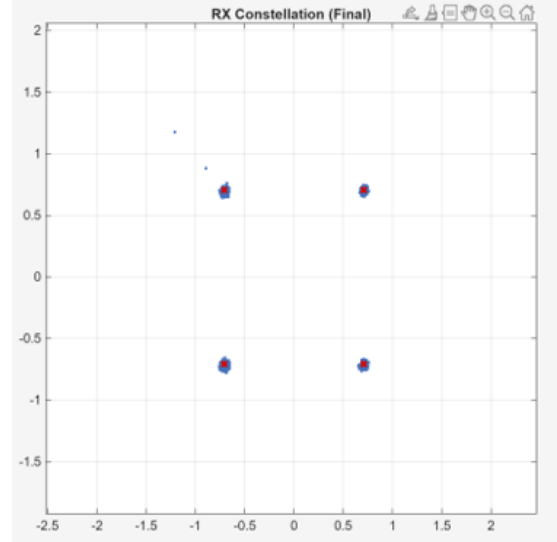
(b) Magnitude Evolution

Figure 14: Channel 5 Characteristics.

To better see the effect of Channel 5 on the transmitted symbols, we observed the received constellation after using the two different training methods we implemented, Block and Comb. While we were transmitting using Block training (the first method we implemented), a portion of the symbols we sent were amplified by the time varying magnitude of the channel, thus resulting in the constellation on the left. However, after the implementation of the Comb training method, which allowed a continuous estimation of the magnitude of the channel, we were able to unveil the cause of the former received constellation by plotting the channel evolution over time, shown in Figure 14b.



(a) PDP



(b) Magnitude Evolution

Figure 15: Channel 5 .

7 Spectral Efficiency

Spectral Efficiency (η) is defined as:

$$\eta = \frac{R}{B} = \frac{N_{sc} \cdot \log_2(M)}{N_{sc} + N_{CP}} \quad [\text{bit/s/Hz}] \quad (6)$$

For our default configuration we have $N_{sc} = 512$, $N_{CP} = 256$, QPSK, so:

$$\eta = \frac{512 \cdot 2}{512 + 256} = \frac{1024}{768} \approx 1.33 \text{ bit/s/Hz} \quad (7)$$

Improving spectral efficiency requires modifying system parameters such as the modulation order M , the guard interval N_{cp} , or the ratio of user data to frame overhead. The following analysis explores these potential improvements alongside the physical constraints of the acoustic channel.

7.1 Increasing Modulation Order (M)

One of the most direct methods to improve efficiency is upgrading the modulation scheme from QPSK ($M = 4$) to higher orders like 16-QAM ($M = 16$). Since 16-QAM encodes 4 bits per subcarrier compared to 2 bits for QPSK, this modification would double the numerator in the efficiency formula. For the current system configuration, this would raise the efficiency to:

$$\eta(16\text{-QAM}) = \frac{512 \times 4}{768} \approx 2.66 \text{ bit/s/Hz}$$

However, this improvement is strictly limited by the available Signal-to-Noise Ratio. Higher-order constellations pack symbols closer together in the complex plane, making them more susceptible to noise.

7.2 Reducing Cyclic Prefix Length (N_{cp})

The current system utilizes a Cyclic Prefix of 256 samples, which is 50% of the useful symbol length. Reducing this overhead significantly increases the data rate. For example, shortening the CP to 128 samples would reduce the symbol duration in the denominator, resulting in a theoretical efficiency of:

$$\eta = \frac{1024}{512 + 128} \approx 1.6 \text{ bit/s/Hz}$$

This represents an approximate 20% increase in efficiency. The limitation of this approach is the risk of Inter-Symbol Interference.

8 Real World Transmission

8.1 Throughput Maximization

The physical throughput R of an OFDM system is given by the ratio of useful bits per symbol to the total symbol duration:

$$R = \frac{N_{sc} \cdot \log_2(M)}{T_{total}} = \frac{N_{sc} \cdot \log_2(M) \cdot B}{N_{sc} + N_{cp}} \quad (8)$$

Where:

- N_{sc} is the number of active subcarriers (512).
- M is the modulation order (4 for QPSK, so 2 bits/symbol).
- B is the system bandwidth.
- N_{cp} is the length of the Cyclic Prefix (overhead).

From this equation, it is clear that to maximize R , we must either maximize the bandwidth B or minimize the overhead N_{cp} . We tested the system by modifying these two key parameters. Increasing the modulation order, as discussed before, could also be a solution, but we decided not to proceed in this direction as the transmission would have been much less resistant to noise.

8.1.1 Bandwidth Expansion

Our default configuration used a conservative bandwidth of $B = 2$ kHz centered at 8 kHz. To push the data rate, we expanded the bandwidth to 16 kHz. We shifted the carrier frequency to $f_c = 12$ kHz. This placed the signal spectrum from 4 kHz to 20 kHz, utilizing almost the entire available audio range while avoiding near-DC interference and the Nyquist cutoff.

8.1.2 Minimizing Overhead (Cyclic Prefix)

While the CP protects against Inter-Symbol Interference, it consumes time that could be used for data. We defined three experimental scenarios to test the result reducing progressively the CP length:

Table 1: Experimental results for Throughput Maximization.

Bandwidth	Cyclic Prefix (N_{cp})	Throughput	BER
16 kHz	128	25,600 bps	6.76 %
16 kHz	32	30,118 bps	9.95 %
16 kHz	0	32,000 bps	13.78 %

The experimental data presented above demonstrates a direct trade-off between throughput and link reliability. With a guard interval of 128 samples, the system maintained a BER of 6.76%.

Aggressively reducing the Cyclic Prefix to just 32 samples boosted the effective data rate to 30.1 kbps while keeping the error rate at 9.95%, just below the 10% threshold. This result effectively identifies the system's performance limit.

Eliminating the overhead entirely maximized the raw throughput to 32 kbps, but the link quality degraded significantly to 13.78%. This failure confirms that without any guard interval, the tail of the impulse response from previous symbols spills into the current symbol, causing Inter-Symbol Interference.

8.2 Image Transmission

We successfully implemented an image transmission on the real audio channel. The source image was first converted to grayscale and resized to 128×128 pixels to keep the transmission time reasonable. Each pixel (valued 0-255) was serialized into an 8-bit binary sequence, resulting in a total payload of $128 \times 128 \times 8 = 131,072$ bits. This structured data is more sensitive to errors than random noise; a single bit flip in the Most Significant Bit (MSB) of a pixel can turn a black pixel white, creating visible noise.

8.2.1 Dynamic Frame Adaptation

Unlike previous experiments with fixed transmission lengths, this task required adapting the frame duration to the payload. The system calculates the exact number of OFDM symbols needed by dividing the total bit count by the capacity of a single symbol ($N_{sc} \times \log_2 M$). Any remaining space in the final symbol is filled with Zero Padding to ensure the data perfectly matches the IFFT block size; these padding bits are simply stripped by the receiver prior to image reconstruction.

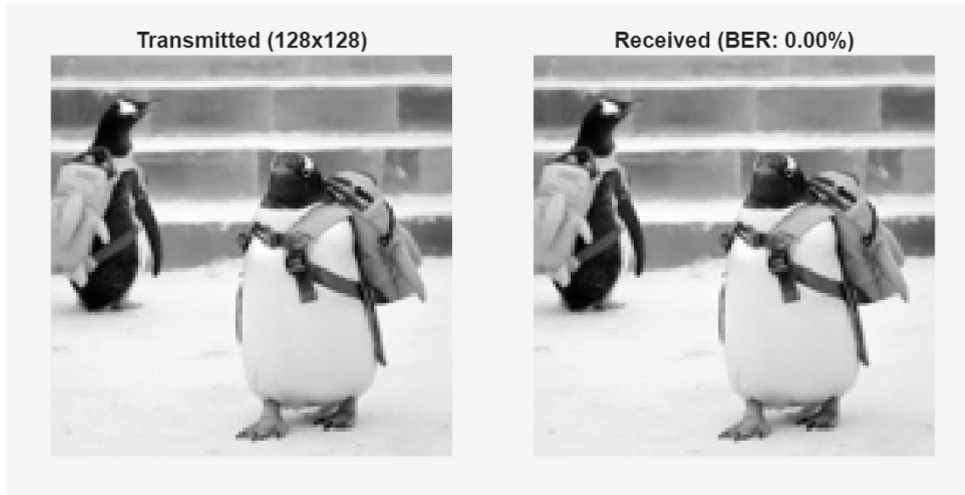


Figure 16: Transmitted and Received Image (0% BER).

As shown in Figure 16 we successfully transmitted the image in the audio channel, the 0% BER reception confirms the robustness of our system.

8.3 Receiver Diversity (MRC)

To further improve the robustness of our acoustic link, we implemented Receiver Diversity. This technique utilizes two different microphones, the built-in laptop microphone and the external USB microphone to capture the transmitted signal simultaneously.

We used the Maximal Ratio Combining (MRC) algorithm to merge these two independent signal paths. Unlike simple averaging, MRC weights the received signals (Y_1, Y_2) according to the complex conjugate of their respective channel estimates (H_1, H_2). The combined symbol estimate $\hat{S}$ is calculated as:

$$\hat{S} = \frac{H_1^* Y_1 + H_2^* Y_2}{|H_1|^2 + |H_2|^2} \quad (9)$$

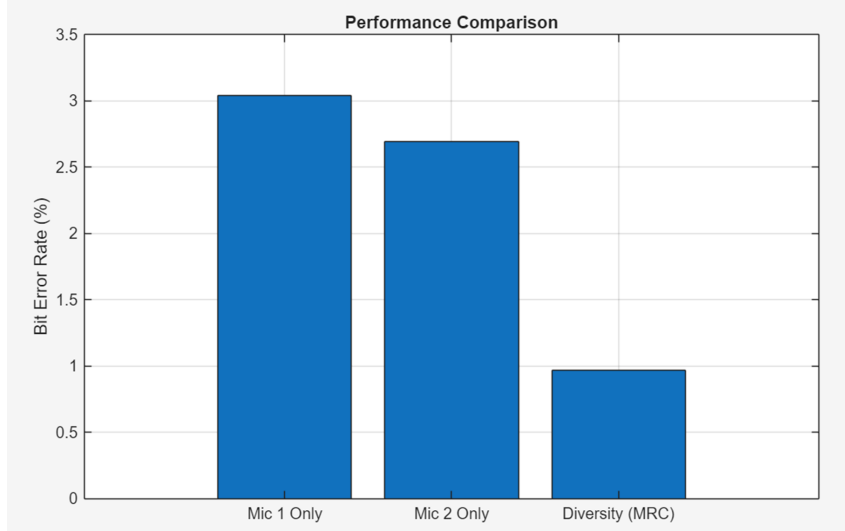


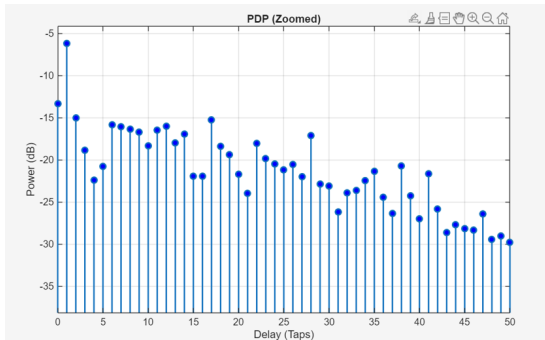
Figure 17: Performance comparison of Single vs Dual microphone setups.

The results, presented in Figure 17, demonstrate the clear advantage of this approach. When decoding using only a single microphone, the system had a BER of approximately 3.0% (Mic 1) and 2.7% (Mic 2). However, using MRC, the BER decreased to less than 1.0%. This improvement confirms that spatial diversity effectively mitigates multipath fading.

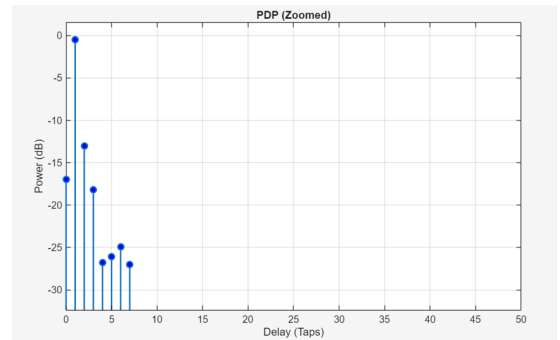
8.4 Environmental Comparison

We tested the system in two distinct locations, to highlight the differences in the estimated channels:

1. Narrow Hallway: The PDP showed dense, long-lasting echoes. The BER was higher due to strong reverberation and noise.
2. Outdoor: The PDP was much shorter (fewer reflections). The system achieved 0% BER easily due to the lack of multipath interference.



(a) Narrow hallway channel



(b) Outdoor channel

Figure 18: Environmental comparison.

9 Conclusion

The implementation of this acoustic OFDM transceiver provided valuable insights into the practical constraints of digital communication systems beyond ideal simulations.

Our analysis of channel estimation strategies demonstrated that Block estimation is effective only in environments where the channel doesn't change much during the transmission. In the presence of time varying channels, this method failed. So we implemented the Viterbi algorithm to keep track of the phase drift. This was successful, but still showed a high BER on Channel 2. Consequently, Comb estimation was identified as the necessary solution, as its continuous pilot insertion allows for robust tracking of phase and amplitude variations throughout the transmission.

Regarding throughput optimization, we established a trade-off between data rate and robustness against Inter-Symbol Interference. By expanding the signal bandwidth and reducing the Cyclic Prefix to a minimum of 32 samples, we increased the throughput from approximately 3 kbps to over 30 kbps. However, the complete removal of the guard interval resulted in immediate link failure, confirming that a minimum Cyclic Prefix is strictly necessary.

Finally, the implementation of Receiver Diversity using MRC significantly reduced the Bit Error Rate, proving that spatial diversity is an effective technique for mitigating multipath fading. Overall, the project confirmed that OFDM is a robust modulation scheme for acoustic transmission, provided that the system parameters are properly optimized for the specific channel conditions.